

$$= (0.03 \times 10^{-3})(3.00 \times 10^{-3}) / (3 \times 10^{-3}) = 0.03 \times 10^{-3} \text{ J.}$$

- 2 B** Shortest path is perpendicularly across. Hence resultant velocity should be in this direction. To “eliminate” the downstream effect, $40 \sin \theta = 10$, which gives θ of 14° .
Shortest time requires that the perpendicular velocity be largest. Hence $\theta = 0$.

- 3 A** The speed of the projectile should be decreasing to a non zero minimum on reaching the top of its